EBEC EMPOWERED
BEST ENGINEERING COMPETITION

BEST PORTO

“;DROP TABLE TEAMS;--

CONTENTS

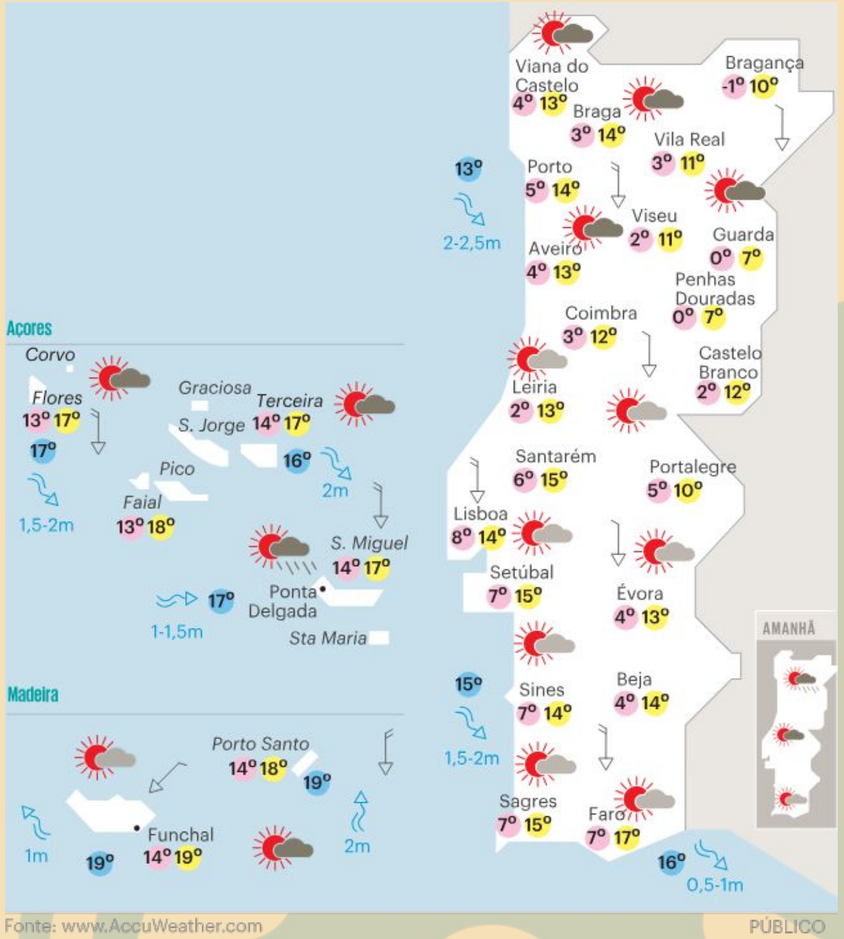
1. Introduction and Problem Statement
2. Level 1-5
3. Interface
4. Conclusion

Introduction

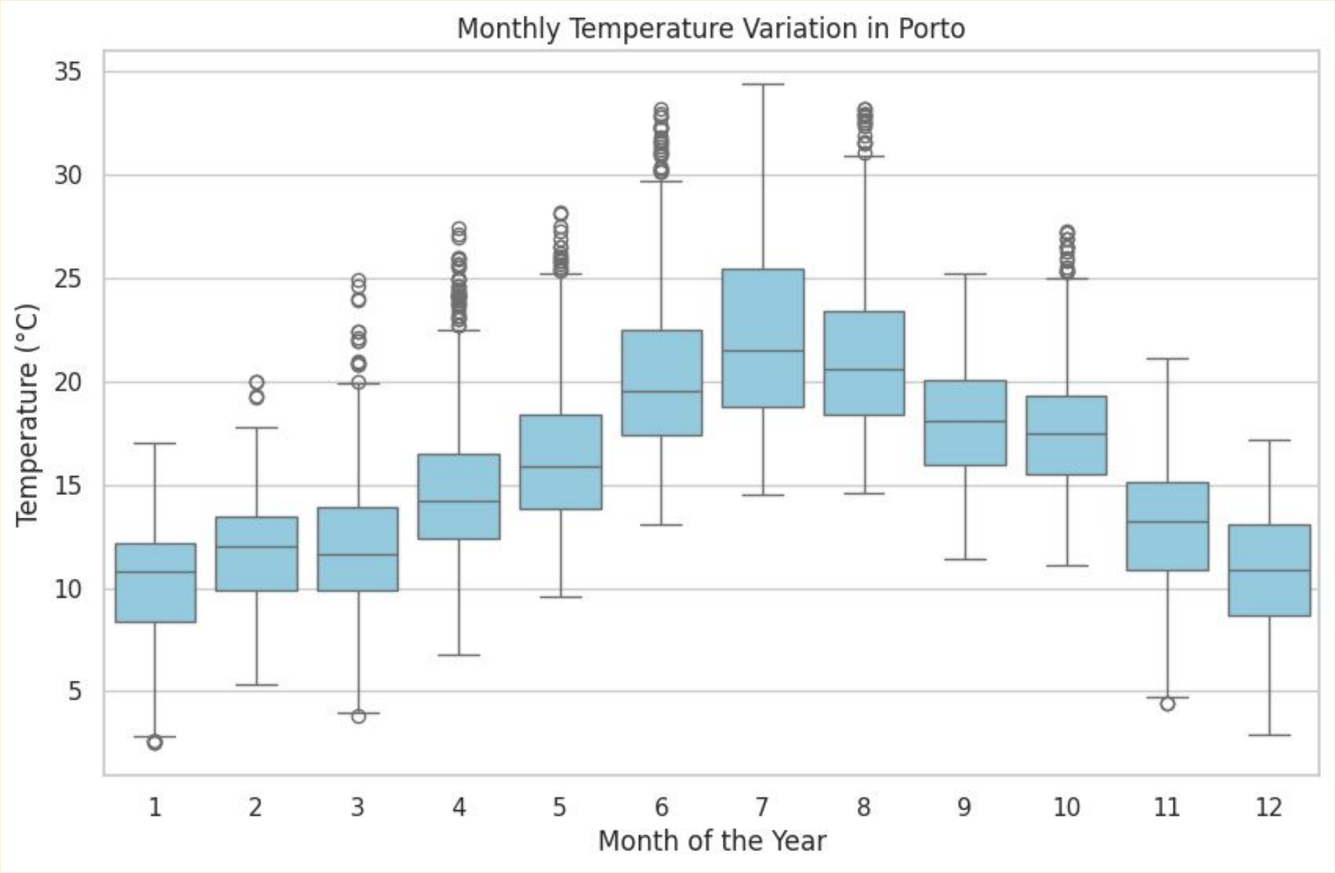
The Problem: Weather is a chaotic system. Sudden changes in rain, fog, and temperature significantly increase accident risks and disrupt traffic.

Our Goal: To explore if Machine Learning can meaningfully support meteorological prediction and highway safety management for Grupo Brisa.

Our Approach: A 6-level development pipeline, moving from basic classification to real-world operational impact and future forecasting.



Preprocessing and Feature Engineering



Level 1: Rain Classification

Goal: Binary classification (Rain: True/False)

Model: RandomForestClassifier

The Challenge: Class Imbalance (Many more dry days than rainy days)

Technique: Random Undersampling

Results:

	Without undersampling	With undersampling
Accuracy	92.7%	91.0%
F1	80.7%	91.0%

Level 2: Temperature Forecasting

Goal: Predict the exact numerical temperature value

Model: XGBRegressor

Mean absolute error: <0.003

Level 3: Unsupervised Snow Detection

Goal: Pattern discovery and anomaly detection for rare snow events

Model: HDBScan (High-Performance Density-Based Clustering).

➔ Unlike K-Means, it handles noise and identifies clusters of varying shapes/densities

The Challenge: No labeled data available; the model must identify structure on its own

Feature Logic: Clustered data based on Temperature, Humidity, and Dew Point

Level 4: Weather-Driven Accidents

Goal: Predict the exact number of emergency vehicles needed (3 cars per accident)

Model: XGBoost

Data Integration & Refining: Merged weather and accident datasets by Date and District, keeping only unique daily entries per location and aggregating hourly weather into daily means and rainy hour counts.

Level 5: Meteorology Forecasting

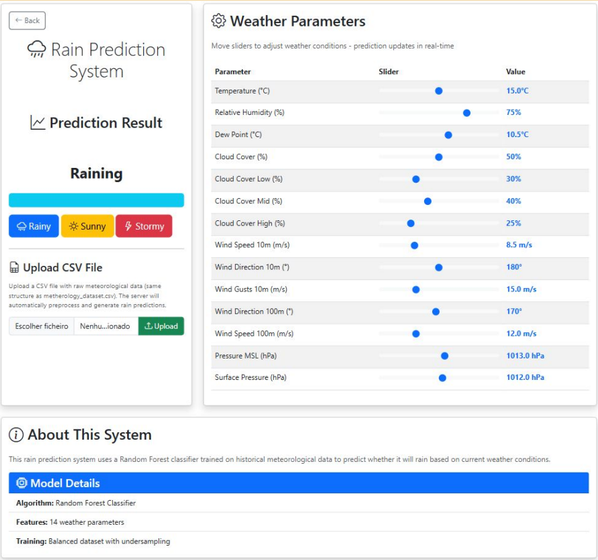
Goal: Proactive prediction of Rain, Temperature, and Cloud Cover for the next 24 hours.

Model: RandomForestClassifier/XGBoost

The Challenge: Cloud cover proved highly unpredictable; after testing multiple regression models, **XGBoost** was selected for its superior ability to handle non-linear variance.

Normalization: Switched to MinMaxScaler to stabilize the diverse ranges of future-targeted features.

Interface


← Back
Rain Prediction System
Prediction Result
Raining
Rainy
Sunny
Stormy
Upload CSV File
Upload a CSV file with raw meteorological data (same structure as metherology_dataset.csv). The server will automatically preprocess and generate rain predictions.
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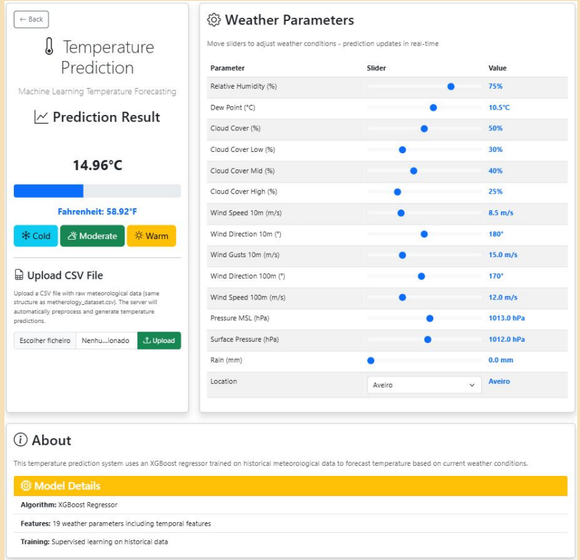
Nenhu...ionado
Upload
Weather Parameters
Move sliders to adjust weather conditions - prediction updates in real-time
Parameter
Slider
Value
Temperature (°C) 15.0°C
Relative Humidity (%) 75%
Dew Point (°C) 10.5°C
Cloud Cover (%) 50%
Cloud Cover Low (%) 30%
Cloud Cover Mid (%) 40%
Cloud Cover High (%) 25%
Wind Speed 10m (m/s) 8.5 m/s
Wind Direction 10m (°) 180°
Wind Gusts 10m (m/s) 15.0 m/s
Wind Direction 100m (°) 170°
Wind Speed 100m (m/s) 12.0 m/s
Pressure MSL (hPa) 1013.0 hPa
Surface Pressure (hPa) 1012.0 hPa
About This System
This rain prediction system uses a Random Forest classifier trained on historical meteorological data to predict whether it will rain based on current weather conditions.
Model Details
Algorithm: Random Forest Classifier
Features: 14 weather parameters
Training: Balanced dataset with undersampling


← Back
Temperature Prediction
Machine Learning Temperature Forecasting
Prediction Result
14.96°C
Fahrenheit: 58.92°F
Cold
Moderate
Warm
Upload CSV File
Upload a CSV file with raw meteorological data (same structure as metherology_dataset.csv). The server will automatically preprocess and generate temperature predictions.
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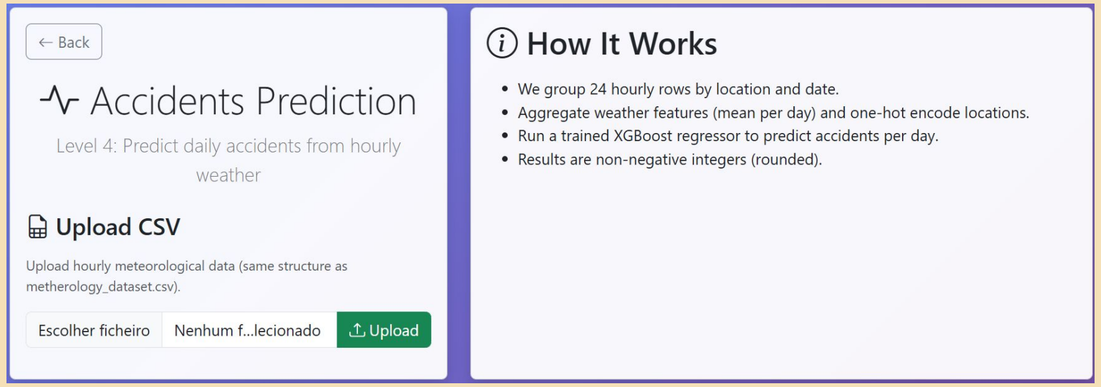
Nenhu...ionado
Upload
Weather Parameters
Move sliders to adjust weather conditions - prediction updates in real-time
Parameter
Slider
Value
Relative Humidity (%) 75%
Dew Point (°C) 10.5°C
Cloud Cover (%) 50%
Cloud Cover Low (%) 30%
Cloud Cover Mid (%) 40%
Cloud Cover High (%) 25%
Wind Speed 10m (m/s) 8.5 m/s
Wind Direction 10m (°) 180°
Wind Gusts 10m (m/s) 15.0 m/s
Wind Direction 100m (°) 170°
Wind Speed 100m (m/s) 12.0 m/s
Pressure MSL (hPa) 1013.0 hPa
Surface Pressure (hPa) 1012.0 hPa
Rain (mm) 0.0 mm
Location
Aveiro
Aveiro
About
This temperature prediction system uses an XGBoost regressor trained on historical meteorological data to forecast temperature based on current weather conditions.
Model Details
Algorithm: XGBoost Regressor
Features: 19 weather parameters including temporal features
Training: Supervised learning on historical data


← Back
Accidents Prediction
Level 4: Predict daily accidents from hourly weather
Upload CSV
Upload hourly meteorological data (same structure as metherology_dataset.csv).
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Upload
How It Works
We group 24 hourly rows by location and date.
Aggregate weather features (mean per day) and one-hot encode locations.
Run a trained XGBoost regressor to predict accidents per day.
Results are non-negative integers (rounded).